F55 — Phase 3 integers: the ~4 real light hadrons confirm substrate-primary integers (vector mesons n_C=5, baryons C_2=6=n_C+1), but the bound-state→integer MECHANISM hits a located wall (dim-vs-Casimir; leading candidate = spin-statistics +1: boson→n_C, fermion→n_C+1). Values DERIVED-as-substrate-primary; mechanism CANDIDATE, honestly not forced.

Lyra (Claude Opus 4.8)

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F55 — Phase 3 integers: partial, and the located wall

0. Result

The deflated Phase 3 (Grace/Elie: ~200 → 4 statistically-real light hadrons) gives clean substrate-primary integers, confirmed:

state	constituents	spin	integer = m/(π^{n_C}·m_e)	substrate value
proton	qqq	½	6.000	C_2 = 6 = n_C+1
neutron	qqq	½	6.008	C_2 = 6
ρ	q$\bar{q}$	1	4.958	n_C = 5
ω	q$\bar{q}$	1	5.005	n_C = 5

The values ARE substrate primaries (n_C for vector mesons, C_2 = n_C+1 for baryons) — this confirms the volume-cell picture (mass = integer·π^{n_C}·m_e, the integer = number of substrate volume-cells the state wraps). The proton is exact to four decimals (6.000); the rest are structural (~1%). Good news: the surviving signal points at the two smallest primaries, not a scatter of large integers.

But the bound-state → integer MECHANISM is not cleanly derived. I hit a real wall, and I report it precisely rather than fake a unification.

1. The wall: dim-vs-Casimir (the integers aren't one invariant)

The natural first guess — "the integer = the dimension of the bound-state's substrate K-type" — works for one and breaks for the other: - Vector meson (spin 1): a spin-1 object is the substrate vector K-type V_(1,0), whose dimension is n_C = 5. So vector meson integer = dim(vector rep) = n_C. ⍰ Clean. - Baryon (spin ½): integer = 6 = C_2. But SO(5) has no dimension-6 representation (dims are 1, 4, 5, 10, 14, 16, …). So 6 is not a dimension — it's the Casimir C_2 (= Casimir of the adjoint V_(1,1)). So the meson integer is a *dimension* and the baryon integer is a *Casimir* — not the same invariant. The "integer = K-type dimension" reading fails for the baryon.

That is the located wall: the two integers are both substrate-primary, but they are *different kinds* of invariant (dim vs Casimir), so there is no single "integer = (one invariant of the bound-state K-type)" law yet.

2. Leading candidate: spin-statistics +1 (boson → n_C, fermion → n_C+1)

The cleanest pattern that fits both, flagged as CANDIDATE (K231c — not asserted):

$$\text{integer} = n_C + \begin{cases} 0 & \text{integer spin (vector meson, boson)} \\ 1 & \text{half-integer spin (baryon, fermion)} \end{cases}$$

- Vector mesons (bosons, spin 1): n_C = 5.
- Baryons (fermions, spin ½): n_C + 1 = 6 = C_2.

The +1 tracks the fermionic/half-integer nature — and C_2 = n_C + 1 is an established substrate identity, so the fermion integer is "the boson integer plus the spin-statistics cell." Substrate-mechanism candidate: a fermion (half-integer K-type, the spinor double-cover) wraps one extra substrate volume-cell relative to the corresponding boson — the Z_2/spin-statistics cell. This is suggestive and *unifies* the two integers (both = n_C + [statistics]), but I have not derived why the fermion wraps exactly +1 cell. It is the candidate, not the theorem.

(Alternative readings I decline to force: baryon C_2 = rank·N_c (color, 2·3=6) — a different substrate form of 6, the Casey #5 multiplicity; or baryon = adjoint V_(1,1) Casimir. The spin-statistics +1 reading is preferred because it *unifies* with the meson via n_C, but the multiplicity of "6" is itself a flag — don't pick without a mechanism.)

3. Honest tiering

- DERIVED (value level): the four real light-hadron integers are substrate primaries — vector mesons n_C = 5, baryons C_2 = n_C+1 = 6. Confirms the volume-cell picture (F52). Proton exact (6.000); ρ/ω/n structural (~1%).
- CANDIDATE (mechanism): integer = n_C + [0 boson / 1 fermion]; the +1 = spin-statistics cell (fermion wraps one extra cell). Unifies both integers via n_C; not yet derived why exactly +1.
- WALL (located, honest): the "integer = K-type dimension" law fails (meson = dim n_C, baryon = Casimir C_2, not one invariant). The spin-statistics reading is the candidate bridge; deriving the +1 from the spinor/Z_2 structure is the open mechanism.
- OPEN: why a fermionic (half-integer, spinor double-cover) bound state wraps n_C+1 cells while a bosonic one wraps n_C. → joint with Keeper (Casey #14 spinor/restriction structure) + Elie (verify the +1 across more light hadrons if any survive base-rate).
- Tier: F55 v0.1 Phase 3 — integers confirmed substrate-primary; mechanism = spin-statistics +1 CANDIDATE; dim-vs-Casimir wall located.

4. Handoff + closure

Phase 3's deflated target (derive two small integers) is half-closed: the values are confirmed substrate-primary (volume-cell picture holds, F52 corroborated), and there's a clean candidate mechanism (boson→n_C, fermion→n_C+1=C_2 via a spin-statistics cell) that unifies them. The honest wall: the two integers are different invariants (dim vs Casimir), so "integer = dimension" fails, and the +1 spin-statistics cell — while it unifies and uses the established C_2 = n_C+1 — is not yet derived from the spinor structure. That last step (fermion wraps +1 cell) is the open mechanism, jointly with Keeper (spinor/Casey #14) and Elie (numerical, if more light hadrons survive). @Elie — the unit 156.38 MeV and the n_C / n_C+1 integers are confirmed; the derive-then-check order holds. @Keeper — the +1 = spinor/Z_2 cell is a candidate K-audit item if the mechanism derives.

The discipline note, because it's the weekend's theme: I could have asserted "integer = bound-state Casimir" and moved on; it fails for the meson. The honest result is a confirmed value-level pattern, a unifying candidate mechanism, and a clearly-located wall — derived where derived, candidate where candidate, walled where walled.

— Lyra, Sun 2026-06-07 10:55 EDT. F55 Phase 3 integers: 4 real light hadrons (Grace/Elie deflation from ~200) confirm SUBSTRATE-PRIMARY integers — vector mesons (ρ,ω) → n_C=5, baryons (p,n) → C_2=6=n_C+1 (proton exact 6.000, rest ~1% structural). Volume-cell picture (F52) corroborated. WALL located: "integer = K-type dimension" works for meson (dim V_(1,0) = n_C=5) but FAILS for baryon (6 is not an SO(5) dim — dims are 1,4,5,10,14,16; 6 = C_2 Casimir). Meson uses dim, baryon uses Casimir — not one invariant. LEADING CANDIDATE (unifies, K231c-flagged): integer = n_C + [0 boson / 1 fermion]; vector meson (spin 1) → n_C, baryon (spin ½) → n_C+1=C_2; the +1 = spin-statistics/spinor cell (C_2=n_C+1 established identity). NOT derived why exactly +1 (open mechanism: fermion wraps one extra cell, joint Keeper spinor/Casey#14 + Elie numerical). Declined alternatives (rank·N_c, adjoint Casimir) — Casey #5 multiplicity of 6, don't force. Tier: values DERIVED-substrate-primary; mechanism CANDIDATE (spin-statistics +1); wall (dim-vs-Casimir) located honestly.